Discrete Structures I

Relations

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1 Introduction

The most direct way to express a relationship between elements of two sets is to use ordered pairs made up of two related elements. For this reason, sets of ordered pairs are called binary relations. In this section we introduce the basic terminology used to describe binary relations. Later in this chapter we will use relations to solve problems involving communications networks, project scheduling, and identifying elements in sets with common properties.

Definition 1. *Let A and B be sets. A binary relation from A to B is a subset of $A \times B$.*

In other words, a binary relation from A to B is a set R of ordered pairs where the first element of each ordered pair comes from A and the second element comes from B.We use the notation $a\ R\ b$ to denote that $(a, b) \in R$ and $a \not R\ b$ to denote that $(a, b) \notin R$. Moreover, when (a, b) belongs to R, a is said to be related to b by R. Binary relations represent relationships between the elements of two sets.

Example 1. *Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$. Then $\{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B. This means, for instance, that $0Ra$, but that $1\not Rb$.*

2 Relations on a Set

Relations from a set A to itself are of special interest.

Definition 2. *A relation on a set A is a relation from A to A.*

In other words, a relation on a set A is a subset of $A \times A$.

Example 2. *Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?*
Because (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b, we see that :
$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$.

Example 3. *Consider these relations on the set of integers :*

1. $R_1 = \{(a, b)|a \leq b\}$
2. $R_2 = \{(a, b)|a > b\}$
3. $R_3 = \{(a, b)|a = bora = -b\}$,
4. $R_4 = \{(a, b)|a = b\}$
5. $R_5 = \{(a, b)|a = b + 1\}$
6. $R_6 = \{(a, b)|a + b \leq 3\}$

Which of these relations contain each of the pairs $(1, 1), (1, 2), (2, 1), (1, -1)$, and $(2, 2)$?
Remark : *Unlike the relations in the previous examples these are relations on an infinite set.*
The pair $(1, 1)$ is in R_1, R_3, R_4, and R_6 ; $(1, 2)$ is in R_1 and R_6 ; $(2, 1)$ is in R_2, R_5, and R_6 ; $(1, -1)$ is in R_2, R_3, and R_6 ; and finally, $(2, 2)$ is in R_1, R_3, and R_4.

3 Properties of Relations

There are several properties that are used to classify relations on a set. We will introduce the most important of these here.

Definition 3. *A relation R on a set A is called reflexive if $(a, a) \in R$ for every element $a \in A$.*

Example 4. *Consider the following relations on $\{1, 2, 3, 4\}$:*

— $Q_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$
— $Q_2 = \{(1, 1), (1, 2), (2, 1)\}$
— $Q_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$
— $Q_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$
— $Q_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$
— $Q_6 = \{(3, 4)\}$

Which of these relations are reflexive ?

The relations Q_3 and Q_5 are reflexive because they both contain all pairs of the form (a, a), namely, $(1, 1), (2, 2), (3, 3)$, and $(4, 4)$. The other relations are not reflexive because they do not contain all of these ordered pairs. In particular, Q_1, Q_2, Q_4, and Q_6 are not reflexive because $(3, 3)$ is not in any of these relations.

Example 5. *Which of the relations from Example 3 are reflexive ?*
The reflexive relations from Example 3 are R_1 (because $a \leq a$ for every integer a), R_3, and R_4. For each of the other relations in this example it is easy to find a pair of the form (a, a) that is not in the relation. (This is left as an exercise for the reader.)

Example 6. *Is the "divides" relation on the set of positive integers reflexive ?*
Because a|a whenever a is a positive integer, the "divides" relation is reflexive. (Note

that if we replace the set of positive integers with the set of all integers the relation is not reflexive because by definition 0 *does not divide 0.)*

Definition 4. *A relation R on a set A is called symmetric if $(b,a) \in R$ whenever $(a,b) \in R$, for all $a,b \in A$.*
A relation R on a set A such that for all $a,b \in A$, if $(a,b) \in R$ and $(b,a) \in R$, then $a = b$ is called antisymmetric.

Remark : Using quantifiers, we see that the relation R on the set A is symmetric if :

$$\forall a \ \forall b \ \ (\ (a,b) \ \in R \ \longrightarrow \ (b,a) \ \in \ R \ )$$

.
Similarly, the relation R on the set A is antisymmetric if :

$$\forall a \forall b \ (\ ((a,b) \ \in \ R \ \wedge \ (b,a) \ \in \ R \ ) \ \longrightarrow \ (a = b))$$

.
That is, a relation is symmetric if and only if a is related to b implies that b is related to a.
A relation is antisymmetric if and only if there are no pairs of distinct elements a and b with a related to b and b related to a. That is, the only way to have a related to b and b related to a is for a and b to be the same element. The terms symmetric and antisymmetric are not opposites, because a relation can have both of these properties or may lack both of them. A relation cannot be both symmetric and antisymmetric if it contains some pair of the form (a,b), where $a \neq b$.

Example 7. *Which of the relations from Example 4 are symmetric and which are antisymmetric ?*

Solution : The relations Q_2 and Q_3 are symmetric, because in each case (b,a) belongs to the relation whenever (a,b) does. For Q_2, the only thing to check is that both $(2,1)$ and $(1,2)$ are in the relation. For Q_3, it is necessary to check that both $(1,2)$ and $(2,1)$ belong to the relation, and $(1,4)$ and $(4,1)$ belong to the relation. The reader should verify that none of the other relations is symmetric. This is done by finding a pair (a,b) such that it is in the relation but (b,a) is not. Q_4, Q_5, and Q_6 are all antisymmetric. For each of these relations there is no pair of elements a and b with $a \neq b$ such that both (a,b) and (b,a) belong to the relation.

Example 8. *Which of the relations from Example 3 are symmetric and which are antisymmetric ?*
Solution : The relations R_3, R_4, and R_6 are symmetric. R_3 is symmetric, for if $a = b$ or $a = -b$, then $b = a$ or $b = -a$. R_4 is symmetric because $a = b$ implies that $b = a$. R_6 is symmetric because $a + b \leq 3$ implies that $b + a \leq 3$. The reader should verify that none of the other relations is symmetric. The relations R_1, R_2, R_4, and R_5 are antisymmetric. R_1 is antisymmetric because the inequalities $a \leq b$ and $b \leq a$ imply that $a = b$. R_2 is antisymmetric because it is impossible that $a > b$ and $b > a$. R_4 is antisymmetric, because two elements are related with respect to R_4 if and only if they are equal. R_5 is antisymmetric because it is impossible that $a = b + 1$ and $b = a + 1$.

Example 9. *Is the "divides" relation on the set of positive integers symmetric ? Is it antisymmetric ?*
Solution : This relation is not symmetric because $1|2$, but $2 \nmid 1$. It is antisymmetric, for if a and b are positive integers with $a|b$ and $b|a$, then $a = b$.

Definition 5. *A relation R on a set A is called transitive if whenever $(a,b) \in R$ and $(b,c) \in R$, then $(a,c) \in R$, for all a, b, $c \in A$.*

Remark : Using quantifiers we see that the relation R on a set A is transitive if we have $\forall a \forall b \forall c(((a,b) \in R \wedge (b,c) \in R) \longrightarrow (a,c) \in R)$.

Example 10. *Which of the relations in Example 3 are transitive ?*
Solution : The relations R_1, R_2,R_3, and R_4 are transitive. R_1 is transitive because $a \leq b$ and $b \leq c$ imply that $a \leq c$. R_2 is transitive because $a > b$ and $b > c$ imply that $a > c$. R_3 is transitive because $a = \pm b$ and $b = \pm c$ imply that $a = \pm c$. R_4 is clearly transitive, as the reader should verify. R_5 is not transitive because $(2,1)$ and $(1,0)$ belong to R_5, but $(2,0)$ does not. R_6 is not transitive because $(2,1)$ and $(1,2)$ belong to R_6, but $(2,2)$ does not.

Definition 6. *Is the "divides" relation on the set of positive integers transitive ?*
Solution : Suppose that a divides b and b divides c. Then there are positive integers k and l such that $b = ak$ and $c = bl$. Hence, $c = a(kl)$, so a divides c. It follows that this relation is transitive.

Definition 7. *A relation on a set A is called an equivalence relation if it is reflexive, symmetric, and transitive.*

Definition 8. *Let R be the relation on the set of integers such that aRb if and only if $a = b$ or $a = -b$. We showed that R is reflexive, symmetric, and transitive. It follows that R is an equivalence relation.*

Definition 9. *A relation R on a set S is called a partial ordering or partial order if it is reflexive, antisymmetric, and transitive. A set S together with a partial ordering R is called a partially ordered set, or poset, and is denoted by (S,R). Members of S are called elements of the poset.*

Example 11. *Show that the "greater than or equal" relation ($\geq$) is a partial ordering on the set of integers.*
Solution : Because $a \geq a$ for every integer a, $\geq$ is reflexive. If $a \geq b$ and $b \geq a$, then $a = b$. Hence, $\geq$ is antisymmetric.
Finally, $\geq$ is transitive because $a \geq b$ and $b \geq c$ imply that $a \geq c$.
It follows that $\geq$ is a partial ordering on the set of integers and $(Z,\geq)$ is a poset.